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IDEAS TODAY - IMPACT TOMORROW

Minor Project Ideas

Python & Java

20 detailed, ready-to-build project ideas with features, tech stack and extensions

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Welcome

A minor project is your first chance to turn classroom concepts into something real. This guide by **Logic Legend** gives you 10 Python and 10 Java project ideas, each with an overview, key features, tech stack, difficulty, estimated time and an extension idea to make your project stand out.

What's inside

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1. Python Project Ideas	Data, AI/ML, automation, desktop and web apps	PY-01 to PY-10
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Quick comparison: Python or Java?

Factor	Python	Java
Best for	AI/ML, data analysis, automation, quick prototypes	Enterprise apps, Android, large structured systems
Learning curve	Gentle, short and readable code	Steeper, strong OOP and type safety
Typical database	SQLite, MySQL, MongoDB	MySQL via JDBC or JPA/Hibernate
Typical interface	Tkinter, Streamlit, Flask	Swing, JavaFX, Spring Boot, Android
Viva strengths	Shows modern, trending skills	Shows solid OOP, DB and architecture skills

Section 1: Python Project Ideas

10 projects spanning AI, data, automation and apps

PY-01 Smart Expense Tracker with Analytics

OVERVIEW	A desktop or web app that records daily expenses, categorises them and shows where the money goes using charts and monthly summaries.
KEY FEATURES	• Add, edit and delete income and expense entries • Category-wise and month-wise charts • Budget limit alerts when spending crosses a threshold • Export reports to CSV or PDF
TECH STACK	Python, Tkinter or Streamlit, SQLite, Pandas, Matplotlib
DIFFICULTY	Easy to Medium | **Estimated time:** 3 to 4 weeks
EXTENSION IDEA	Add receipt scanning with OCR (pytesseract) to auto-fill amounts.

PY-02 AI Chatbot for College Queries

OVERVIEW	A chatbot that answers student questions about admissions, timetables, fees, events and placements using natural language processing.
KEY FEATURES	• Intent classification using a JSON knowledge base • Web chat interface built with Flask • Fallback replies and a 'talk to admin' option • Chat history stored in a database
TECH STACK	Python, NLTK or spaCy, Flask, scikit-learn, HTML/CSS
DIFFICULTY	Medium | **Estimated time:** 4 to 5 weeks
EXTENSION IDEA	Connect it to Telegram or WhatsApp and add voice input.

PY-03 Face Recognition Attendance System

OVERVIEW	Marks attendance automatically by recognising student faces through a webcam, removing manual roll calls.
KEY FEATURES	• Register a new face with multiple samples • Real-time recognition with timestamped attendance • Daily and monthly attendance reports • Admin login to manage students
TECH STACK	Python, OpenCV, face_recognition, SQLite or CSV, Tkinter
DIFFICULTY	Medium to Hard | **Estimated time:** 5 to 6 weeks
EXTENSION IDEA	Add mask detection and email alerts for low attendance.

PY-04 Fake News and Spam Message Detector

OVERVIEW	A machine learning model that classifies news headlines or SMS/e-mail text as genuine or fake/spam, wrapped in a simple web app.
KEY FEATURES	• Text cleaning and TF-IDF feature extraction • Compare Naive Bayes, Logistic Regression and SVM • Confusion matrix and accuracy report • Web form where users paste text and get a verdict
TECH STACK	Python, Pandas, scikit-learn, Flask, Kaggle dataset
DIFFICULTY	Medium | **Estimated time:** 3 to 4 weeks
EXTENSION IDEA	Upgrade to a transformer model such as BERT and compare results.

PY-05 Weather Dashboard and Forecast App

OVERVIEW	Shows current weather, 5-day forecast and air quality for any city using a public weather API.
KEY FEATURES	• City search with auto-suggest • Temperature, humidity and wind graphs • Save favourite cities • Rain alert notifications
TECH STACK	Python, Requests, OpenWeather API, Streamlit, Plotly
DIFFICULTY	Easy | **Estimated time:** 2 to 3 weeks
EXTENSION IDEA	Add a map view and clothing or travel suggestions based on weather.

PY-06 Personal Voice Assistant

OVERVIEW	A desktop assistant that listens to voice commands to open apps, search the web, tell the time, read news and set reminders.
KEY FEATURES	• Speech-to-text and text-to-speech • Wikipedia and web search by voice • Open applications and websites • Reminder and to-do management
TECH STACK	Python, SpeechRecognition, pyttsx3, wikipedia, PyAutoGUI
DIFFICULTY	Easy to Medium | **Estimated time:** 3 weeks
EXTENSION IDEA	Add a wake word and smart-home control through a microcontroller.

PY-07 Student Performance Predictor

OVERVIEW	Predicts whether a student is likely to pass or score high based on attendance, study hours, past marks and other factors.
KEY FEATURES	• Data cleaning and exploratory analysis • Regression and classification models • Feature importance visualisation • Dashboard for teachers to view at-risk students
TECH STACK	Python, Pandas, scikit-learn, Seaborn, Streamlit
DIFFICULTY	Medium | **Estimated time:** 4 weeks
EXTENSION IDEA	Add personalised study recommendations for each student.

PY-08 Price Tracker with Web Scraping

OVERVIEW	Tracks product prices on e-commerce sites and sends an e-mail when a price drops below the target set by the user.
KEY FEATURES	• Scrape product name, price and rating • Store price history in a database • Price-history line chart • Automatic e-mail alert through SMTP
TECH STACK	Python, BeautifulSoup, Requests, SQLite, smtplib, schedule
DIFFICULTY	Medium | **Estimated time:** 3 to 4 weeks
EXTENSION IDEA	Support multiple websites and predict the best time to buy.

PY-09 Secure Password Manager

OVERVIEW	A vault that stores website credentials in encrypted form behind a single master password.
KEY FEATURES	• AES encryption of stored passwords • Strong password generator • Password strength checker • Search, copy-to-clipboard and auto-lock
TECH STACK	Python, cryptography (Fernet), SQLite, Tkinter
DIFFICULTY	Medium | **Estimated time:** 3 weeks
EXTENSION IDEA	Add two-factor authentication and breach checking.

PY-10 Sentiment Analysis of Product Reviews

OVERVIEW	Analyses customer reviews or tweets and reports whether public opinion is positive, negative or neutral.
KEY FEATURES	• Data collection from CSV or API • VADER or TextBlob sentiment scoring • Word clouds and trend charts • Summary report of top positive and negative keywords
TECH STACK	Python, Pandas, NLTK, VADER, WordCloud, Matplotlib
DIFFICULTY	Medium | **Estimated time:** 3 weeks
EXTENSION IDEA	Add aspect-based analysis (battery, price, camera, etc.).

Section 2: Java Project Ideas

10 projects spanning management systems, networking, Spring Boot and Android

JV-01 Online Banking and ATM Simulation

OVERVIEW	A secure banking application that lets customers create accounts, deposit, withdraw, transfer money and view statements.
KEY FEATURES	• Customer and admin login with PIN protection • Deposit, withdrawal and fund transfer • Mini statement and transaction history • Account lock after repeated wrong PIN attempts
TECH STACK	Java, Swing or JavaFX, JDBC, MySQL
DIFFICULTY	Medium | **Estimated time:** 4 to 5 weeks
EXTENSION IDEA	Add OTP verification and interest calculation.

JV-02 Hospital Management System

OVERVIEW	Manages patients, doctors, appointments, billing and medical records in a single application.
KEY FEATURES	• Patient registration and search • Doctor scheduling and appointment booking • Bill generation with printable invoice • Role-based login for admin, doctor and receptionist
TECH STACK	Java, JavaFX or Swing, JDBC, MySQL, iText
DIFFICULTY	Medium | **Estimated time:** 5 weeks
EXTENSION IDEA	Add prescription history and e-mail appointment reminders.

JV-03 Library Management System

OVERVIEW	Automates book issue, return, fine calculation and member management for a library.
KEY FEATURES	• Add, search and categorise books • Issue and return with due dates • Automatic fine calculation • Reports of most borrowed books
TECH STACK	Java, Swing, JDBC, MySQL or SQLite
DIFFICULTY	Easy to Medium | **Estimated time:** 3 to 4 weeks
EXTENSION IDEA	Add barcode scanning and a student-facing web portal.

JV-04 Online Quiz and Examination Portal

OVERVIEW	A web platform where teachers create timed quizzes and students attempt them with instant results.
KEY FEATURES	• Teacher panel to add questions and set timers • Randomised questions and auto-submit on timeout • Instant score and answer review • Leaderboard and result export
TECH STACK	Java, Servlets/JSP or Spring Boot, MySQL, Bootstrap
DIFFICULTY	Medium | **Estimated time:** 4 to 5 weeks
EXTENSION IDEA	Add anti-cheating tab-switch detection and question banks by topic.

JV-05 Multi-User Chat Application

OVERVIEW	A real-time chat system where multiple clients connect to a central server for group and private messaging.
KEY FEATURES	• Client-server communication through sockets • Multithreaded server handling many users • Private and group chat rooms • File sharing and online-user list
TECH STACK	Java, Socket programming, Multithreading, Swing/JavaFX
DIFFICULTY	Medium | **Estimated time:** 3 to 4 weeks
EXTENSION IDEA	Add message encryption and emoji support.

JV-06 Inventory and Billing System

OVERVIEW	Helps a small shop track stock, generate bills and view daily sales.
KEY FEATURES	• Product and supplier management • Barcode-style quick billing • Low-stock warnings • PDF invoice and daily sales report
TECH STACK	Java, JavaFX, JDBC, MySQL, iText or JasperReports
DIFFICULTY	Medium | **Estimated time:** 4 to 5 weeks
EXTENSION IDEA	Add GST calculation and cloud backup.

JV-07 Student Management Portal (Spring Boot)

OVERVIEW	A full-stack portal to manage student records, courses, marks and attendance through REST APIs.
KEY FEATURES	• CRUD operations with Spring Data JPA • Login using Spring Security • Course and marks management • REST API documented with Swagger
TECH STACK	Java, Spring Boot, Spring Security, JPA/Hibernate, MySQL, Thymeleaf
DIFFICULTY	Medium to Hard | **Estimated time:** 5 to 6 weeks
EXTENSION IDEA	Add a React front end and JWT-based authentication.

JV-08 Movie or Event Ticket Booking System

OVERVIEW	Users browse shows, pick seats on a visual seat map and book tickets with a booking confirmation.
KEY FEATURES	• Show listing with date, time and price • Interactive seat selection with real-time availability • Booking history and cancellation • Admin panel to add shows and theatres
TECH STACK	Java, Spring Boot, MySQL, HTML/CSS/JavaScript
DIFFICULTY	Medium to Hard | **Estimated time:** 5 to 6 weeks
EXTENSION IDEA	Add a payment gateway sandbox and QR-code tickets.

JV-09 Employee Payroll System

OVERVIEW	Calculates salaries including allowances, deductions and tax, and produces payslips.
KEY FEATURES	• Employee records and attendance • Automated salary and tax computation • Payslip generation in PDF • Monthly payroll summary reports
TECH STACK	Java, Swing, JDBC, MySQL, iText
DIFFICULTY	Medium | **Estimated time:** 3 to 4 weeks
EXTENSION IDEA	Add leave management and e-mail delivery of payslips.

JV-10 Android Expense and To-Do Manager

OVERVIEW	A mobile app that combines daily task planning with expense tracking and offline storage.
KEY FEATURES	• Add tasks with reminders • Expense entry with categories and graphs • Offline database storage • Dark mode and home-screen widget
TECH STACK	Java, Android Studio, Room database, MPAndroidChart
DIFFICULTY	Medium | **Estimated time:** 4 to 5 weeks
EXTENSION IDEA	Add Firebase sync so data is available across devices.

Section 3: How to Choose and Present

Make your minor project count

Choosing the right project

- **Match your skill level.** Pick a project you can finish in the available time, then add one extension for extra marks.
- **Solve a real problem.** Projects that help a college, shop or community are easier to justify in a viva.
- **Use a real database.** Even a simple SQLite or MySQL backend makes a project feel complete.
- **Build one standout feature.** Charts, PDF reports, e-mail alerts or an ML model give your project an edge.

Suggested project timeline

Week	Milestone
Week 1	Finalise idea, gather requirements, draw use-case and ER diagrams
Week 2	Design database and UI wireframes, set up the repository
Weeks 3 to 4	Build core modules and test each as you go
Week 5	Add the standout feature, handle errors and validate inputs
Week 6	Testing, report writing, PPT and demo practice

Documentation checklist

- Abstract, problem statement and objectives
- Existing system vs proposed system
- System requirements (hardware and software)
- DFD, ER diagram, class and use-case diagrams
- Screenshots, test cases and results
- Conclusion and future scope
- GitHub link with a clear README

Viva tips

- Be ready to explain every line of code and why you chose the tools you did.
- Prepare a short live demo with sample data and a backup screen recording.
- Know your project's limitations and how you would improve it.

Build it. Break it. Improve it.

Good luck from Logic Legend - Ideas Today, Impact Tomorrow.